



Himalayan Glacial Disaster — DRR Concept Diagram

Support options drawing on Japanese technology, following the 25–26 August 2026 Bhote Koshi–Trishuli flood in Nepal (working draft)

Drafted 2026-08-30 · The source reconstruction is still being revised (this sheet shows the best reading as of 2026-08-30 together with its revision history) · Search ongoing; figures provisional · Not an official position of any agency

The source mechanism is still unsettled.
But the 6–10 hours downstream certainly existed — and did not turn into evacuation.

1

What happened

08:37

The glacier terminus fails together with its bedrock (~5,200 m elevation, ~600 m wide, 1,200 m drop). The M5.2 signal **was produced by the collapse**. It enters the Lhende River and turns **directly into a debris flow**

08:50
09:20

Two stage gauges **stop transmitting** (= destroyed). 09:05 DHM becomes aware (**16 min after the wave crossed the border**)

09:15

Mass SMS alert: **679,295 messages**. Only 4 districts targeted

11:45

A second collapse on the same slope (M4.2). Search teams are entering the valley — **no warning is issued**

15:20
16:00

Reaches Devghat → peak **6.57 m** (no second peak)

About 10 hours from collapse to the lowest reach · 675 dead, 2,418 missing

This chronology changed three times in four days. First USGS report → IRDR (previous-day collapse + 18–19 h of impoundment) → satellite analysis (direct debris flow, no evidence of a dam). Because there are no instruments upstream, everything rests on inference from remote data. A GLOF struck the same corridor in July 2025 (11 dead).

2

Diagnosis — deaths concentrated where lead time was longest

District (upstream → downstream)

Margin after alert

Bodies recovered

Rasuwa

-25 min

13

Nuwakot

+5 min

51

Dhading

~1.5–2 h

45

Gorkha

~3–4 h

57

Tanahun

~4–5 h

35

Chitwan

6 h 05 min

246

Nawalparasi E
not alerted

~7–8 h

165

Nawalparasi W
not alerted

~8–9 h

63

A decisive caveat: “bodies recovered” is where a body was found, not where the person died. Depending on how bodies carried down from upstream are treated, the policy conclusion can invert. That is exactly why the verification survey (A-0) must precede any equipment donation. What does not move: 320 people — about 47% of the total — were recovered in the four districts that were never alerted (Gorkha, Tanahun, Nawalparasi East and West).

3

Four blind spots

GAP 1 Four days on, what happened at the source is still unsettled

First USGS report → IRDR (previous-day collapse + prolonged impoundment) → satellite analysis (direct debris flow): **three versions in four days**. Views diverge because there are no upstream instruments and everything is remote inference. That, in itself, is the case for putting monitoring there.

GAP 2 Monitoring watches only for a rise in water

The Rasuwagadhi gauge (446.22) showed no prior change. The three upstream stations **fell silent before reaching alert stage**, and that silence — itself the evidence of destruction — was logged as “missing data”. Neither a falling stage nor missing data is treated as an anomaly.

GAP 3 “A flood has occurred” does not move people

670,000 messages went out at 09:15, yet the wave reached Chitwan **six hours later**. What was needed was not notice that something had happened but “what time it arrives at your village”, followed by tracking updates.

GAP 4 Casualties concentrated among those SMS cannot reach

Of the missing, **933 are tunnel workers** and **589 are foreign nationals** — 63%. Tunnels have neither signal nor escape routes; trekkers have neither Nepali-language SMS nor community networks. Neither group is in the design scope of the existing warning system.

4

The measures as a whole — laid out along the lead-time axis

AT THE SOURCE — THE MECHANISM IS NOT YET SETTLED

B-4 / TRANSBOUNDARY

Peacetime monitoring

Track change in lakes and in unstable slopes, continuously

► ALOS-2/4 · Sentinel Asia

B-1 / DETECT

Collapse detection

Alerts regardless of mechanism — so it holds under either hypothesis

► NIED Hi-net · microbarometers

B-3 / PREDICT — CONDITIONAL

Blockage assessment

May not apply to this event, but the capability is still needed

► Japan’s emergency-survey rules

SOURCE MECHANISM UNRESOLVED — THREE VERSIONS IN FOUR DAYS

First USGS: the 08:37 collapse is the origin → IRDR: a 13:00 collapse the day before, breach after 18–19 h → GFZ and satellite analysis: the 08:37 collapse becomes a debris flow directly; no evidence of a sustained blockage

Orders of magnitude — only the excess volume constrains the source

Frictional melt (even spending all of the 1,200 m drop — 11.8 kJ/kg — caps melt at 3.5% of the ice mass) 10⁵–10⁶ m³

18–19 h of channel blockage and impoundment (a catchment the size of the Lhende) 10⁶ m³

Total volume past Devghat (includes baseflow; cannot constrain the source) 10⁸ m³

↳ of which the excess above baseflow — this is what constrains the source 10⁷ m³

Seismic records for 25 Aug — queried 2026-08-30: nothing found

USGS lists only the two 26 Aug events within 300 km over 8 days (both type=landslide, depth 0 km). Nothing on 25 Aug.

waveform-level check still outstanding

AFTER THE COLLAPSE — 100 KM IN UP TO 10 HOURS (THE ALERT WENT OUT; EVACUATION DID NOT FOLLOW)

B-2 / DISSEMINATE

Automatic sirens

Wired straight through — no human decision in the loop

► Tokachidake wire / vibration sensors

A-2 / DETECT

Track the wave with gauges

Trigger on rate of rise; missing data = maximum alert

► Crisis-management stage gauges, 6 → 30

A-1 / PREDICT — TOP PRIORITY

Arrival times, alerts by village

“Arrives 11:40. Reach high ground by 10:30” — then keep updating

► Flood Risk Line · timeline-based DRR

A-3 / DISSEMINATE

Alert extent = hazard extent

Defined by the river corridor, not by administrative district

► CAP-compliant · the L-Alert design

collapse +13 min +43 min +3–5 h +6 h 05 +7–9 h

Bhote Koshi

Trishuli R.

Narayani R. → India

Rasuwa 13

Nuwakot 51

Dhading 45

Gorkha 57

Tanahun 35

Chitwan 246

Nawalparasi E 165

W 63

▲ Bodies recovered (NDRRMA, 2026-08-29 18:30)

BASIN-WIDE, PEACETIME

A-4 Protocols for tunnel workers and foreign nationals (cell broadcast / tunnel safety standards)

C-1 High-ground refuges and escape routes for every settlement

C-3 Hydropower plants as EWS nodes (monitoring obligations secured through loan and insurance conditions)

A-0 Verification survey (settle the source mechanism; census of where and when each death occurred and which alert was received)

Left: how the source reconstruction changed three times in four days (as of 2026-08-30). Right: propagation of the flood wave after the collapse (the horizontal axis is not to scale). The margin available at the source is disputed, but the 6–10 hours downstream exist under every hypothesis. Only three arrival times are measured — Syabrubesi 08:50, Betrawati 09:20, Devghat 15:20; the rest are interpolated. Bodies recovered is not a count by place of death.

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Japanese technology and the bodies that would deliver it

DETECT

NIED / Japan Meteorological Agency

Hi-net, V-net; automatic seismic and infrasound detection of lahars

Stations sit inside Nepal, so the system can be built without waiting for agreement with China. It needs no identification of mechanism, so it works while expert views still diverge.

DETECT

MLIT / Instrument manufacturers

Crisis-management stage gauges; wire and vibration sensors

The “under ¥1 M, five years unpowered, retrofittable” specification adopted after Typhoon Lionrock in 2016 (24 dead in Iwaizumi).

PREDICT

PWRI / ICHARM / MLIT

Flood Risk Line; timeline-based DRR

This disaster itself left measured data with which to calibrate the model.

CROSS-BORDER

JAXA / Sentinel Asia / ADRC

ALOS-2 and -4 (L-band SAR); International Disasters Charter

There is distinct value in Japan observing and distributing neutrally, as a third country.

DELIVER

MIC / mobile network operators

Emergency cell broadcast (ETWS / CB); L-Alert

Broadcasts to every handset in several languages without any subscriber list, and reaches roaming foreign SIMs.

STRUCTURAL

MLIT Sabo Dept. / JICA / JBIC and NEXI

Emergency survey of landslide dams; unmanned construction

A statutory procedure that runs impounded volume → breach discharge → inundation estimate within 24–48 h. Whether it applies here awaits the mechanism being settled.

6

Sequence of implementation

WHEN	WHAT	WHO
0–3 mo	A-0 verification survey — settle the source mechanism; census of place and time of death and of which alert was received. In parallel, satellite monitoring of the unstable mass	Joint academic survey team / JAXA / ADRC
0–1 yr	A-1 arrival-time forecasting; A-3 review of alert coverage; A-4 tunnel and foreign-national protocols	ICHARM / JICA / MIC
1–2 yr	B-1 source detection network (alerts on mass movement regardless of mechanism); B-2 automatic sirens; B-3 transfer of the channel-blockage procedure. Stage gauges to 30+	NIED / MLIT Sabo Dept.
2–5 yr	C-1 high-ground refuges and escape routes; C-2 land-use guidance; C-3 power plants as nodes, with financial conditions	JICA / JBIC and NEXI
ongoing	Build the transboundary data-sharing framework; institutionalise Japan’s observation supply as a neutral third party	MOFA / ICIMOD / ADRC

On equipment dying within three years: require no mains power (five-year battery); use no consumables; duplicate the link (satellite IoT); hold spares and trained repairers locally; and pay against 95%+ annual uptime rather than against delivery.

7

Stated for credibility: what this cannot do

Satellites could not have predicted this ice–rock avalanche

Brittle failure gives little precursory displacement. The role is limited to three things: updating the lake inventory, rapid post-event assessment, and narrowing down unstable slopes.

Check dams cannot stop an event of this size

Capturing a debris flow that travels over 100 km is not realistic. What works is high-ground refuges, training dykes on tributaries, and scour-resistant design of rebuilt structures.

Precipitation products were powerless here

No rain was involved in this event. The design must give the rainfall and glacial paths independent detection chains and share only the downstream dissemination.

QZSS DC Report cannot be the primary path

Nepal sits near the western edge of the service area, and deep valleys may not give sufficient elevation angle. Start with a measured proof of concept on receivability.

Legend (what the colours mean)

- Detection & observation
- Prediction & decision
- Dissemination & alerting
- Action & structures
- Cross-border & institutions
- Human losses

Group A = immediate (within 1 yr) / Group B = medium term (2 yr; where Japan is distinctively strong) / Group C = structures and land use (3–10 yr). Priority = lead time created ÷ cost × capacity to maintain.

Glossary

DHM: Dept. of Hydrology and Meteorology, Nepal
NDRRMA: National Disaster Risk Reduction and Management Authority
GLOF: glacial lake outburst flood
ICIMOD: Int’l Centre for Integrated Mountain Development
ICHARM: Int’l Centre for Water Hazard and Risk Management

CAP: Common Alerting Protocol
ETWS / CB: cell-broadcast emergency alerting
SAR: synthetic aperture radar (sees through cloud)
Unmanned construction: earthworks by remotely operated machines
Timeline DRR: planning backwards from the arrival time

Find it early upstream. Deliver time downstream.

What this disaster lacked was not a means of knowing the danger, but a path from knowing to escaping. Sources: GFZ, ICIMOD, Copernicus / Planet satellite analysis, IRDR, USGS, NDRRMA, MLIT, NIED, JAXA, JICA and others